Projective-plane triangulations yield orderable cographic matroids

Carptopus
carptopus@163.com

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Abstract

Crenshaw and Oxley conjectured that every 3-connected orderable binary matroid is graphic. We give a topological construction that disproves this conjecture. If T is a finite simplicial triangulation of the real projective plane and G is its 1-skeleton, then the cographic matroid $M^*(G)$ is orderable. The ordering is induced by face corners. For every bond of G, the resulting 2-regular adjacency graph is identified with a piecewise-linear separating level set. An Euler-characteristic argument shows that this level set has one component. Starting from the six-vertex triangulation with 1-skeleton K_6 and repeatedly applying stellar subdivision to triangular faces gives infinitely many 3-connected, regular, non-graphic, orderable binary matroids. We also give the explicit K_6 certificate and an exact finite checker as a reproducibility aid.

1 Introduction

An ordering of a matroid assigns a reversible cyclic ordering to every circuit. It is *consistent* when common adjacent pairs are treated compatibly across the circuit family; a matroid admitting such an assignment is *orderable* [1]. Crenshaw and Oxley proved strong structural restrictions on orderable matroids and conjectured that every 3-connected orderable binary matroid is graphic [1, Conjecture 4]. Their theorem for 4-connected regular matroids leaves the exact 3-connected boundary open.

We show that this boundary contains a natural infinite family of counterexamples. The triangular faces of a projective-plane triangulation define a single global adjacency relation on the edges of its 1-skeleton. On each bond this relation has degree two. The main point is connectedness. The bond can be realized as the mid-level set of a two-valued piecewise-linear function on the surface. Because both sides of a bond are connected, the two complementary subsurfaces are connected. The positive Euler characteristic of the real projective plane then forces the common boundary to be one circle.

The six-vertex triangulation makes the smallest member especially concrete: its 1-skeleton is K_6, so $M^*(K_6)$ itself is an orderable, regular, binary, 3-connected, non-graphic matroid. Repeated face subdivisions produce the infinite family.

2 Definitions and a global-adjacency criterion

All graphs and simplicial complexes are finite. For a connected graph G, the circuits of the cographic matroid $M^*(G)$ are the bonds of G, that is, the inclusion-minimal nonempty edge cuts. For a nonempty proper vertex set $S \subset V(G)$, the cut $\delta(S)$ is a bond exactly when both induced subgraphs $G[S]$ and $G[V(G) \setminus S]$ are connected.

We use the following immediate sufficient form of consistency.

Lemma 2.1 (Global adjacency)**.** *Let M be a matroid. Suppose that a simple graph Λ on $E(M)$ has the property that, for every circuit C of M, the induced graph $\Lambda[C]$ is a single cycle. Reading that cycle in either direction gives consistent reversible cyclic orderings of all circuits. In particular, M is orderable.*

Proof. All circuit orderings are restrictions of the same unordered adjacency relation. Hence an adjacent pair shared by two circuits is adjacent in both induced cycles, which is the required consistency. □

3 The projective-plane theorem

Theorem 3.1. *Let T be a finite simplicial triangulation of $\mathbb{RP}^2$, and let $G = T^{(1)}$ be its 1-skeleton. Then $M^*(G)$ is orderable.*

Proof. Define a simple graph Λ_T on $E(G)$ as follows. Two edges of G are adjacent in Λ_T when they are distinct edges of a common triangular face of T. Since T is simplicial, a pair of incident edges belongs to at most one common face, so this is a well-defined global adjacency relation.

Fix a bond $B = \delta(S)$. Every edge $uv \in B$ belongs to exactly two triangular faces. In a face uvw, the third vertex w lies on exactly one side of the cut. Consequently exactly one of uw, vw is another edge of B, and that edge is adjacent to uv in Λ_T. The two faces incident with uv have different third vertices, so they supply two different neighbors. Thus $\Lambda_T[B]$ is 2-regular and hence is a disjoint union of cycles.

It remains to show that there is only one cycle. Define a piecewise-linear map

$$h : |T| \longrightarrow [0, 1]$$

by setting $h(v) = 0$ for $v \in S$, $h(v) = 1$ for $v \notin S$, and extending affinely over every simplex. Put

$$C = h^{-1}(1/2), \qquad N_0 = h^{-1}([0, 1/2]), \qquad N_1 = h^{-1}([1/2, 1]).$$

In every mixed triangular face, C is the segment joining the midpoints of its two crossing edges. These segments join across common edges and form a disjoint union of simple closed curves. Their combinatorial adjacency graph is precisely $\Lambda_T[B]$. Since $1/2$ is not the value of any vertex, the same triangle-by-triangle local model shows that N_0, N_1 are compact surfaces and that $C = \partial N_0 = \partial N_1$.

The subspace N_0 strongly deformation retracts onto the full induced subcomplex $T[S]$. Indeed, in each simplex let p be the sum of the barycentric coordinates at vertices in S. On N_0 we have $p \geq 1/2$. Join each point by the straight-line homotopy to the point obtained by dividing its S-coordinates by p and setting the remaining coordinates to zero. Along this homotopy the sum of the S-coordinates is $(1-t)p + t \geq 1/2$, so the homotopy stays in N_0; the simplexwise formulas agree on common faces. The same argument applies to N_1 and $T[V(G) \setminus S]$. Since B is a bond, the 1-skeleta $G[S]$ and $G[V(G) \setminus S]$ are connected. Therefore N_0 and N_1 are connected compact surfaces with common boundary C.

Let b be the number of components of C. A connected compact surface with b boundary components has Euler characteristic at most $2 - b$, in both the orientable and nonorientable cases. If $b \geq 2$, then $\chi(N_0) \leq 0$ and $\chi(N_1) \leq 0$. Because C is a union of circles, $\chi(C) = 0$, and additivity gives

$$1 = \chi(\mathbb{RP}^2) = \chi(N_0) + \chi(N_1) - \chi(C) \leq 0,$$

a contradiction. Hence $b = 1$. Thus $\Lambda_T[B]$ is one cycle for every bond B, and Lemma 2.1 proves that $M^*(G)$ is orderable. $\square$

4 An infinite family of counterexamples

On vertex set $\{0, 1, 2, 3, 4, 5\}$, take the ten faces

$$013, 014, 024, 025, 035, 123, 125, 145, 234, 345.$$

Every edge occurs in exactly two faces, every vertex link is a 5-cycle, and $6 - 15 + 10 = 1$. This is a triangulation of $\mathbb{RP}^2$ whose 1-skeleton is K_6.

Corollary 4.1. *The matroid $M^*(K_6)$ is 3-connected, regular, binary, non-graphic, and orderable. Hence it is a counterexample to Crenshaw–Oxley Conjecture 4.*

Proof. Theorem 3.1 gives orderability. Cographic matroids are regular and binary. The simple 3-connected graph K_6 has a 3-connected cycle matroid, and duality preserves matroid 3-connectivity. Finally, a cographic matroid of a 3-connected graph is graphic only when the graph is planar; K_6 is nonplanar [2]. $\square$

Start with this triangulation T_0. Given a triangular face abc, insert a new vertex x in its interior and replace abc by abx, bcx, cax. Denote the resulting stellar subdivision by T_{m+1}, and write $G_m = T_m^{(1)}$.

Theorem 4.2. *The matroids $M^*(G_m)$, $m \geq 0$, are infinitely many pairwise nonisomorphic 3-connected, regular, binary, non-graphic, orderable matroids. Every member is a counterexample to Crenshaw–Oxley Conjecture 4.*

Proof. Stellar subdivision preserves the underlying surface and the simplicial triangulation, so Theorem 3.1 applies at every stage. We prove 3-connectivity of G_m by induction. The new vertex x has the three neighbors a, b, c, while every old edge is retained. After deleting at most two vertices, the surviving old graph is connected by the induction hypothesis. If x survives, at least one of a, b, c also survives, so x lies in the same component. Thus G_{m+1} is 3-connected.

Every G_m contains the initial K_6 as a subgraph, and is therefore nonplanar. The standard graphic–cographic planarity criterion, together with 3-connectivity, implies that $M^*(G_m)$ is non-graphic. It is cographic and hence regular and binary, and its matroid 3-connectivity follows from that of G_m. Finally, each subdivision adds three edges, so the ground-set sizes are distinct. $\square$

5 The explicit K_6 certificate and verification boundary

For reproducibility, the six vertex links, written as cyclic orders of the other endpoints, are

$$\begin{array}{c|l} 0 & (1,3,5,2,4) \\ 1 & (0,3,2,5,4) \\ 2 & (0,4,3,1,5) \\ 3 & (0,1,2,4,5) \\ 4 & (0,1,5,3,2) \\ 5 & (0,2,1,4,3). \end{array}$$

Declaring two K_6-edges adjacent when they occur at a common face corner gives 30 global adjacent pairs. The 31 bonds of K_6 split by cut type as six of size 5, fifteen of size 8, and ten of size 9.

The accompanying checker reconstructs the binary cographic representation, all circuits, the global adjacency relation, and matroid connectivity, and verifies that every induced bond adjacency is a single cycle. A second checker performs stellar subdivisions through a finite calibration range.

These computations are not used to prove the general theorem or the infinite family. They are exact positive controls for the smallest example and the implementation. No timeout or failed search is used as mathematical evidence.

6 Scope and prior-work boundary

The result disproves the stated 3-connected binary conjecture, but it does not conflict with the 4-connected regular theorem of Crenshaw and Oxley. Every G_m retains triangles of the initial K_6; such a triangle is a three-element cocircuit of $M^*(G_m)$, and together with 3-connectivity this gives an exact 3-separation. We do not classify all orderable cographic matroids or all surface triangulations that yield them. The Euler-characteristic argument is specific to the positive Euler characteristic of $\mathbb{RP}^2$; it does not assert an analogous theorem on the torus or on higher-genus surfaces.

The literature search checked the published and arXiv versions of [1] and searched directly for the $M^*(K_6)$, cographic, orderable, and projective-plane-triangulation combination. No direct prior statement of the theorem or counterexample was located. This is a documented search boundary, not a claim of exhaustive worldwide priority. The manuscript is an internally verified candidate proof and has not undergone external peer review.

AI-assisted research disclosure

Carptopus is the responsible author and takes responsibility for the claims, proofs, source use, and released artifacts. OpenAI Codex was used for AI-assisted literature organization, proof development and stress testing, exact verification, adversarial auditing, and manuscript preparation.

References

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